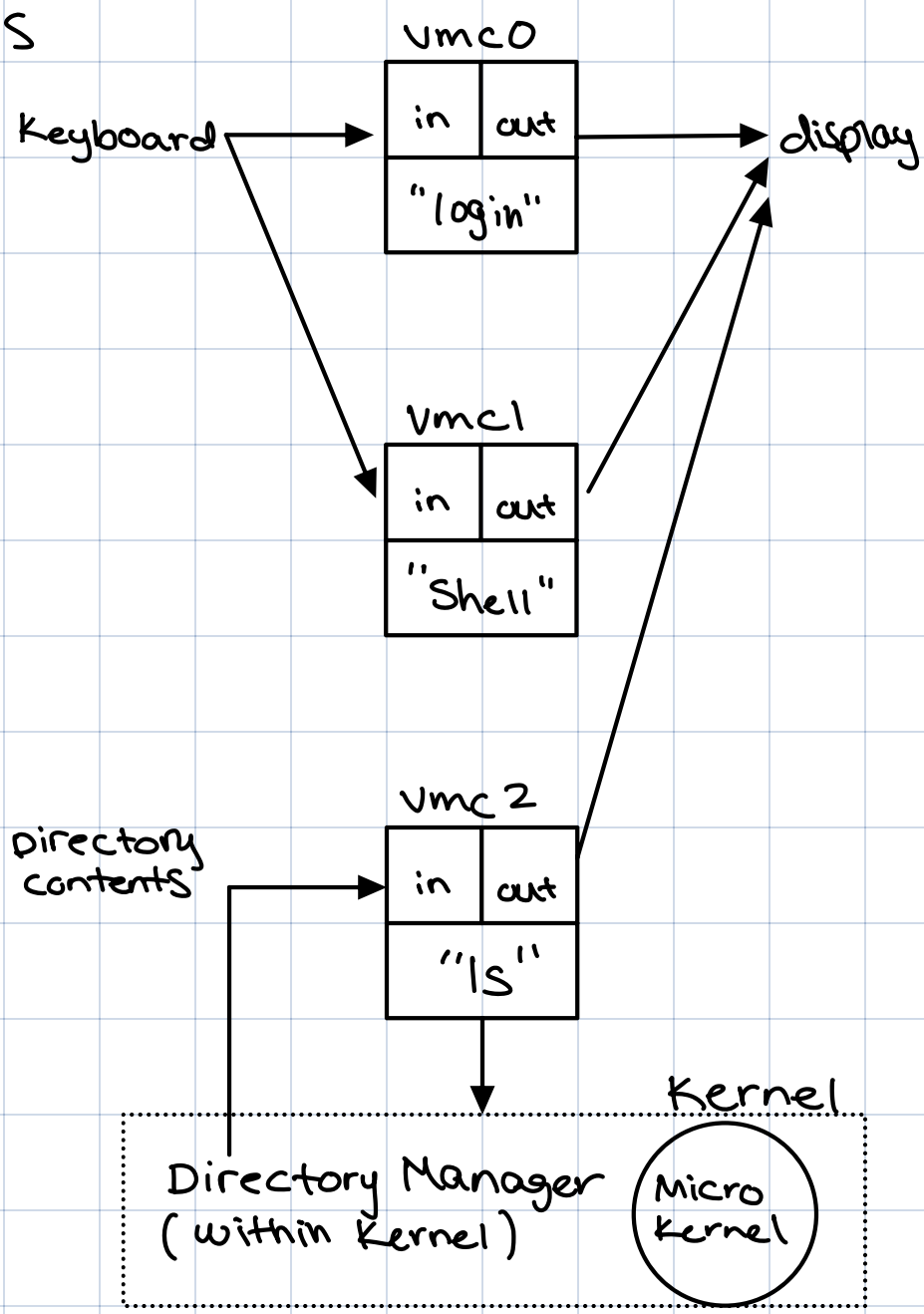
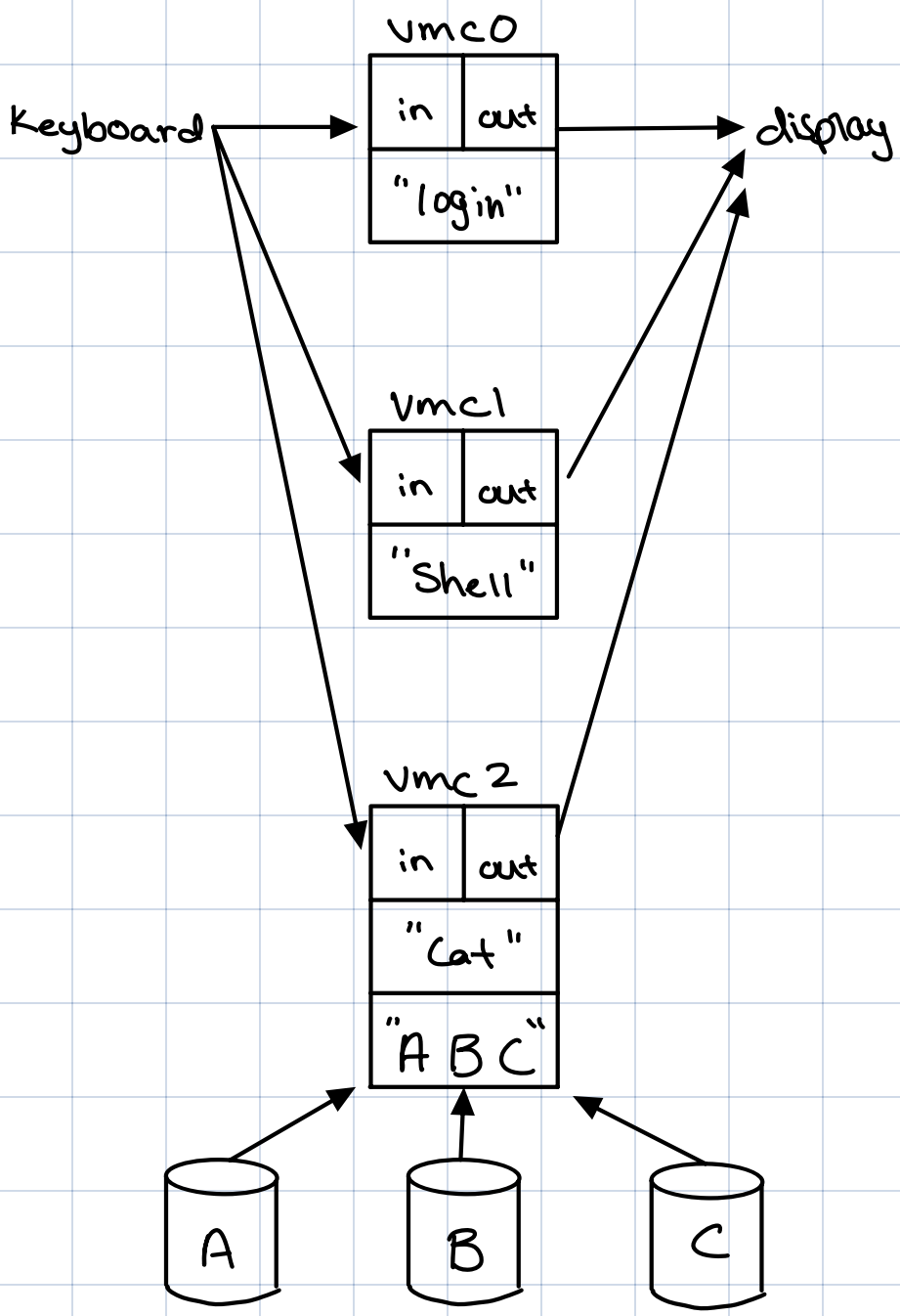
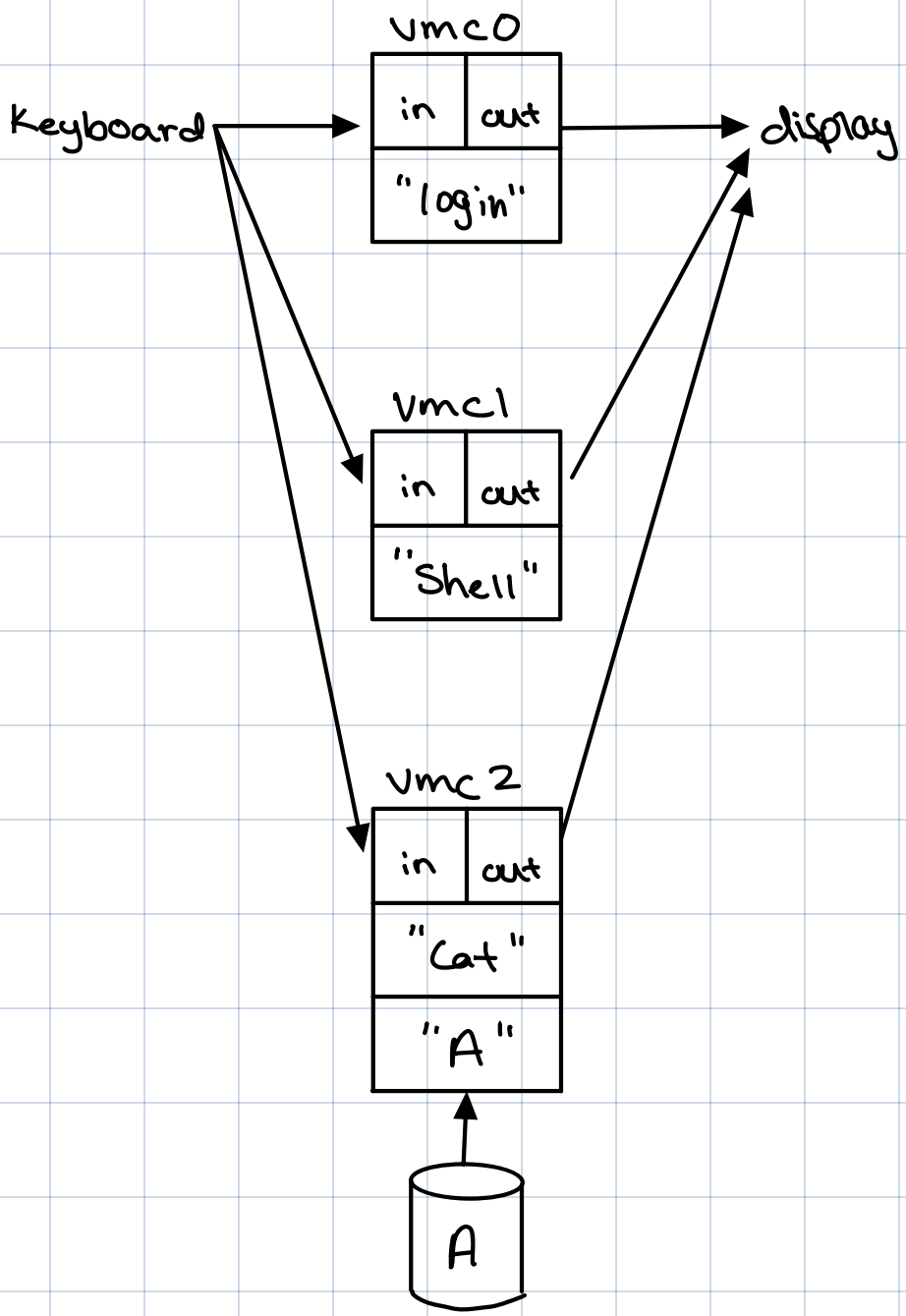
4. ls


vmc0
in
out
"login"
Keyboard
display
vmc1
in
out
"Shell"
vmc 2
in
out
"ls"
Directory contents
Kernel
Directory Manager
(within Kernel)
Micro Kernel

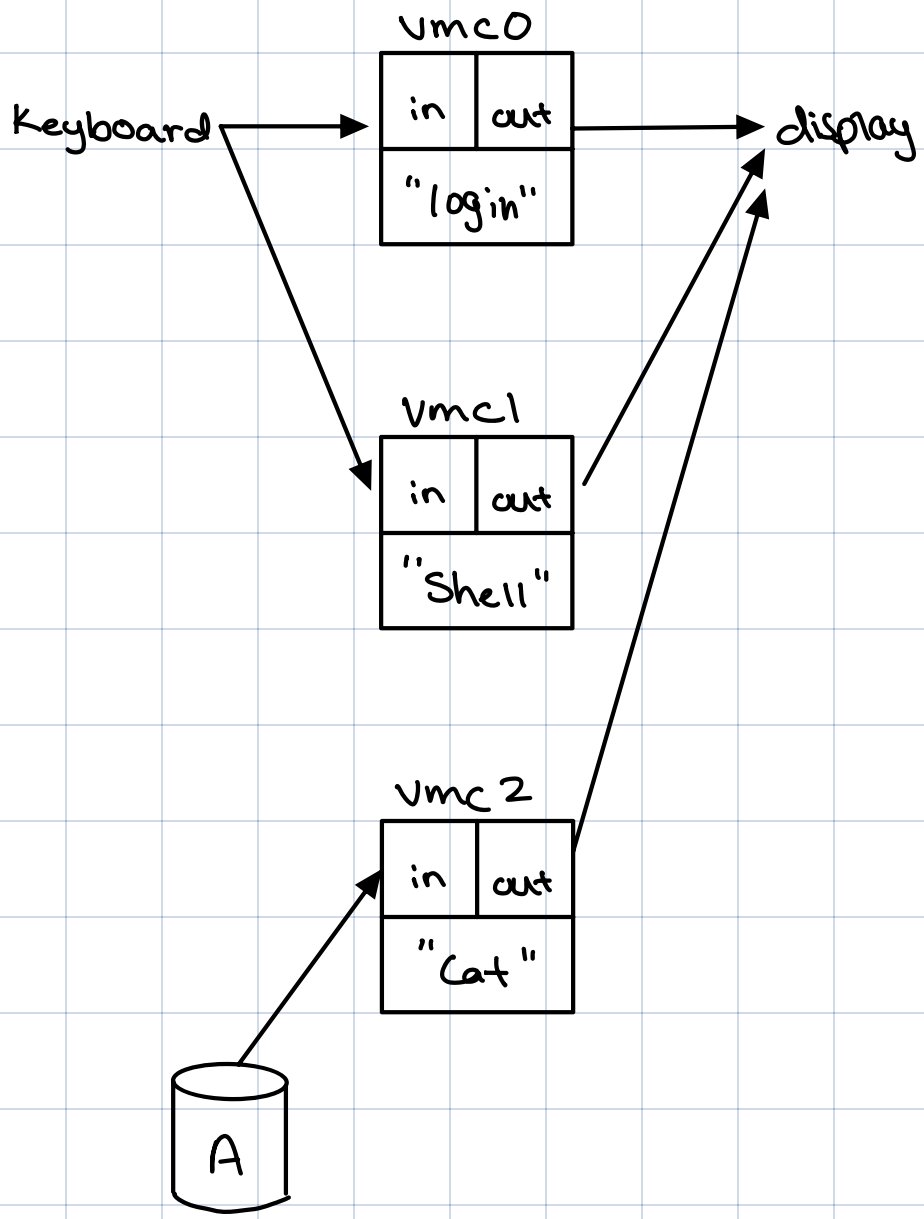
5a. cat A B C


vmc0
in
out
"login"
Keyboard
display
vmc1
in
out
"Shell"
vmc2
in
out
"Cat"
"A B C"
A
B
C

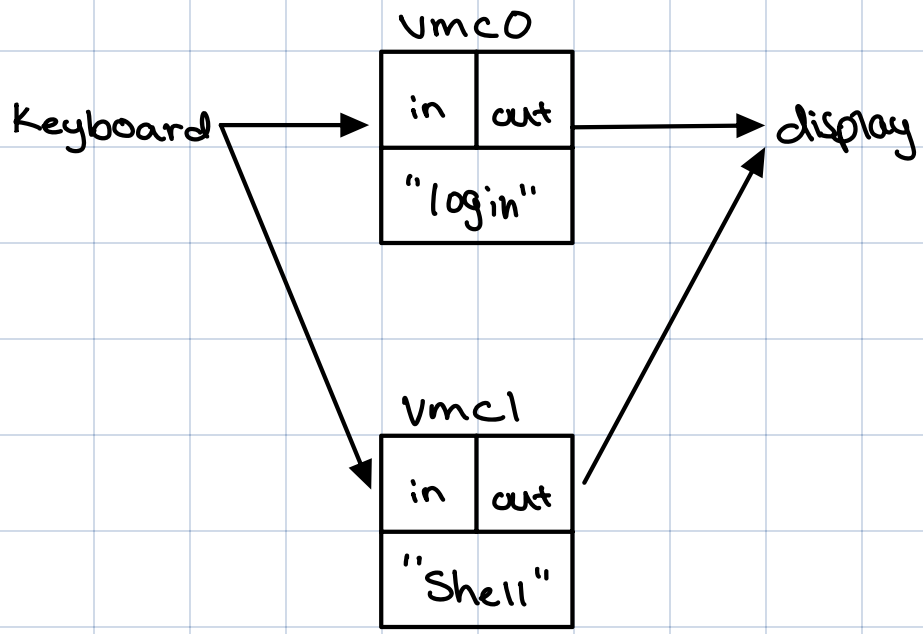
5b. cat A

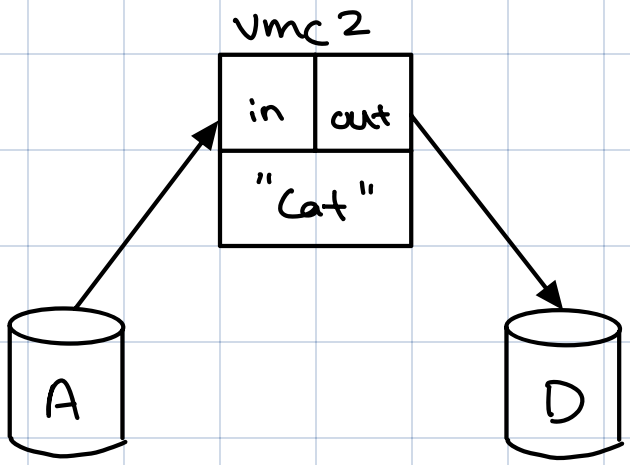

vmc0
in
out
"login"
Keyboard
display
vmc1
in
out
"Shell"
vmc2
in
out
"Cat"
"A"
A

5c. cat < A

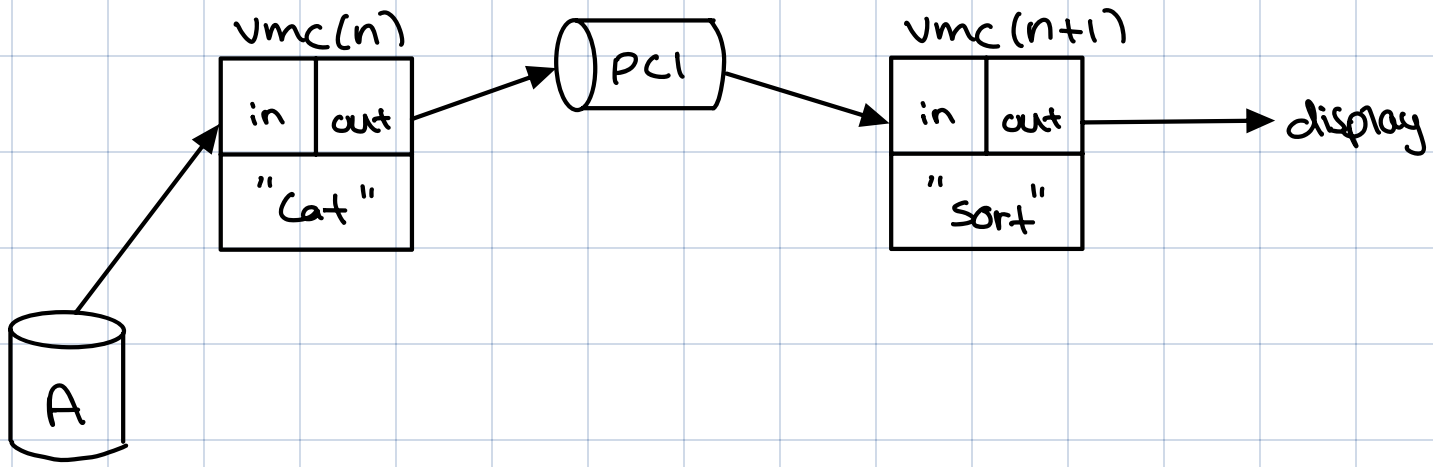

vmc0
in
out
"login"
Keyboard
display
vmc1
in
out
"Shell"
vmc2
in
out
"Cat"
A

5d. cat < A > D

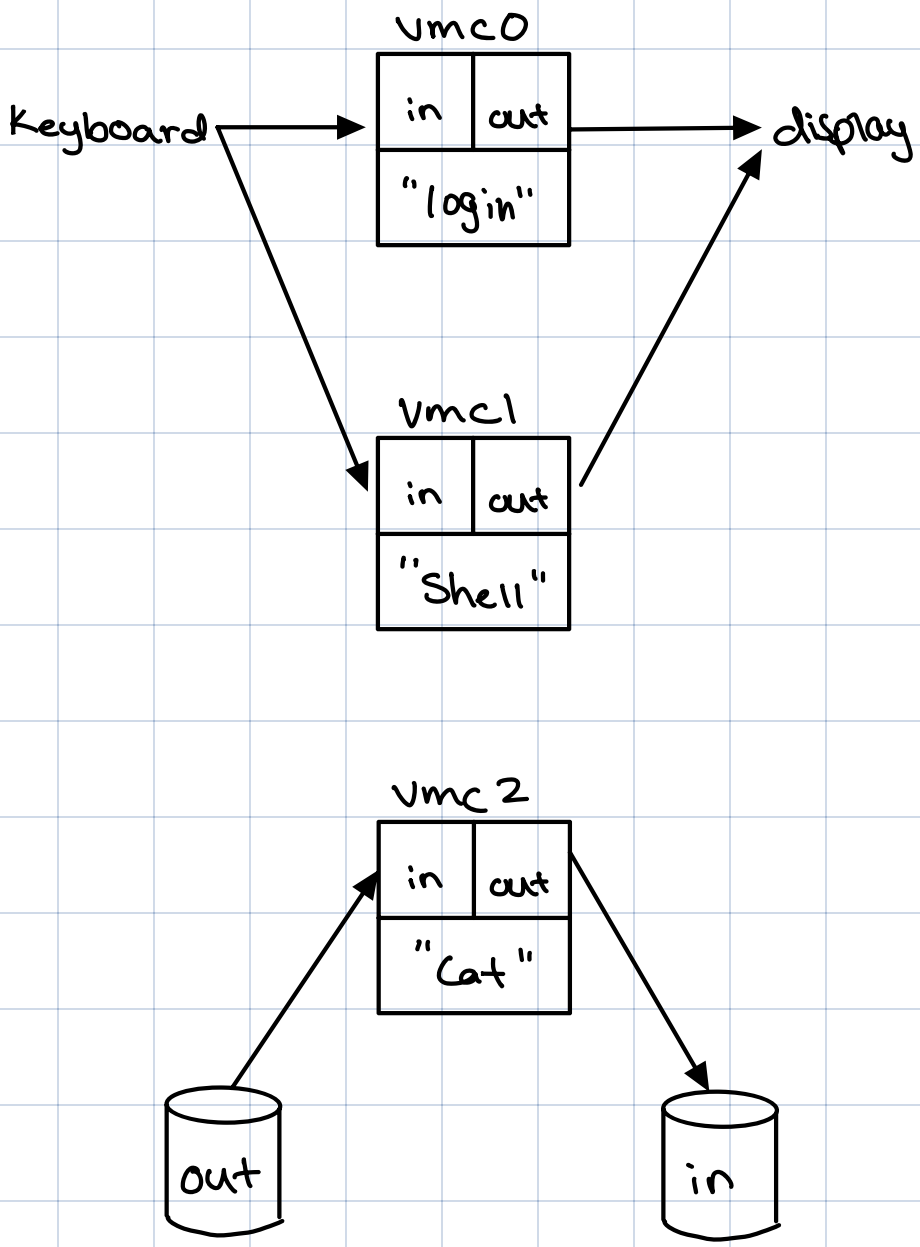




6. cat < A | sort



10. cat < out > in


vmc0
in
out
"login"
Keyboard
display
vmc1
in
out
"Shell"
vmc2
in
out
"Cat"
out
in